

KUOPIO MAANINKA, Finland

WMO: 027880

Lat: 63.143N

Lon: 27.313E

Elev: 300

StdP: 14.54

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.8	-11.0	-22.8	1.6	-17.8	-15.3	2.4	-11.0	20.4	28.4	18.4	26.9	2.2	330	0.627

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.0	80.6	65.8	76.9	64.1	73.5	62.0	68.1	76.9	66.1	73.7	63.8	70.7	5.9	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.2	94.1	71.8	62.9	86.9	69.9	60.8	80.3	67.8	32.6	76.8	31.0	73.4	29.2	70.8	76.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.4	16.7	14.5	DB	-20.8	83.8	10.9	5.5	-28.7	87.8	-35.0	91.0	-41.1	94.1	-49.0	98.1	(4)
(5)			WB	-19.9	70.2	10.9	3.6	-27.8	72.8	-34.2	74.9	-40.4	76.9	-48.3	79.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	39.1	15.2	18.2	25.4	36.6	49.5	57.0	62.7	59.1	50.5	38.8	31.4	23.0		
	DBStd	18.19	13.10	13.38	9.19	6.48	8.18	7.02	6.21	5.57	5.94	6.40	7.75	12.03		
	HDD50	5078	1078	890	764	403	110	17	1	4	66	350	558	837		
	HDD65	9542	1543	1310	1229	853	482	255	119	195	434	812	1008	1302		
	CDD50	1086	0	0	0	1	95	226	393	287	82	2	0	0		
	CDD65	76	0	0	0	0	2	14	47	13	0	0	0	0		
	CDH74	663	0	0	0	0	51	128	388	96	0	0	0	0		
	CDH80	122	0	0	0	0	2	18	80	22	0	0	0	0		
	WSAvg	6.9	6.3	6.9	7.4	6.7	7.1	7.1	6.0	6.1	6.8	7.8	7.2	7.2		
	PrecAvg	25.1	1.7	1.5	1.3	1.3	1.9	2.8	3.5	2.8	2.1	2.2	2.0	2.0		
(15) Precipitation	PrecMax	31.9	3.1	2.9	2.8	2.3	4.8	4.8	6.1	5.9	3.9	3.9	4.5	3.4		
	PrecMin	18.0	0.5	0.1	0.2	0.4	0.5	0.8	0.9	0.9	0.8	0.5	0.3	0.7		
	PrecStd	3.3	0.7	0.7	0.6	0.5	1.1	1.1	1.3	1.0	1.0	0.9	0.9	0.7		
	PrecStd	3.3	0.7	0.7	0.6	0.5	1.1	1.1	1.3	1.0	1.0	0.9	0.9	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.1	40.8	46.8	61.8	79.5	82.6	86.3	83.2	69.8	55.8	49.4	41.7		
		MCWB	36.5	37.3	39.1	48.7	59.5	64.2	68.6	68.9	61.3	51.6	47.4	40.7		
	2%	DB	35.1	37.5	42.1	55.8	74.5	77.8	81.9	75.8	65.2	52.5	45.0	38.1		
		MCWB	34.0	35.1	36.7	46.2	57.9	63.5	67.5	63.7	58.0	49.9	43.5	36.7		
	5%	DB	33.3	34.9	38.7	51.1	69.7	73.6	78.3	72.3	62.1	50.1	42.6	36.1		
		MCWB	32.6	33.8	34.8	43.0	57.1	61.2	65.8	62.1	56.6	47.8	41.6	34.9		
	10%	DB	31.9	33.3	36.6	47.3	64.4	69.8	74.7	69.1	59.5	47.7	40.1	34.4		
		MCWB	31.3	32.4	33.6	40.6	52.7	59.1	64.1	60.9	55.3	45.6	39.3	33.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.0	37.6	39.9	50.9	62.4	67.9	71.6	69.9	61.6	52.8	47.5	41.0		
		MCDB	38.3	40.0	44.3	57.9	74.5	76.3	80.4	80.6	67.7	54.7	48.8	42.0		
	2%	WB	34.4	35.6	37.4	47.2	59.9	65.5	69.2	65.8	59.4	50.7	43.7	36.8		
		MCDB	34.9	37.0	41.1	54.7	70.9	73.9	78.3	72.1	63.1	52.0	44.6	37.6		
	5%	WB	32.9	33.9	35.7	43.7	57.5	62.4	67.5	63.6	57.5	48.0	41.6	35.0		
		MCDB	33.4	34.7	37.9	49.6	69.1	72.1	75.8	69.2	60.8	49.5	42.5	35.9		
	10%	WB	31.4	32.6	34.2	41.2	54.2	60.1	65.6	62.1	55.9	45.7	39.4	33.7		
		MCDB	32.0	33.4	36.0	46.2	63.1	68.3	73.2	67.6	59.1	47.4	40.0	34.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.3	11.1	13.9	14.9	17.2	15.6	16.0	14.5	11.9	8.2	6.1	7.8		
		MCDBR	6.6	8.2	13.8	22.0	26.0	22.1	21.8	20.3	16.3	10.9	8.0	6.3		
		MCWBR	6.4	7.3	10.1	13.7	12.6	10.5	10.2	10.3	10.3	8.6	7.5	6.0		
	5% WB	MCDBR	6.4	7.9	11.8	20.2	24.0	20.6	19.7	16.6	14.2	10.1	7.7	6.3		
		MCWBR	6.3	7.4	9.2	13.1	12.4	10.2	10.2	9.1	9.4	8.8	7.3	6.1		
(40) Clear-Sky Solar Irradiance	taub		0.227	0.284	0.312	0.350	0.347	0.346	0.370	0.352	0.329	0.311	0.244	N/A		
	taud		2.000	2.111	2.182	2.321	2.456	2.483	2.430	2.467	2.521	2.391	2.090	N/A		
	Ebn at Noon		123	201	246	263	276	278	266	262	246	198	115	N/A		
	Edh at Noon		10	21	29	32	30	30	31	28	22	16	9	N/A		
(44) All-Sky Solar Radiation	RadAvg		57	233	713	1149	1516	1666	1586	1210	677	261	65	25		
	RadStd		8	54	82	140	165	111	164	132	76	46	12	3		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (36 neighbors)		+1.01	N/A	N/A	N/A	N/A	N/A	-310	N/A	N/A	N/A

Nomenclature: See separate page